PROJECT PROPOSAL

Hardware Acceleration Analysis for Edge AI: A Comparative Study of Hailo AI Accelerator and CPU Performance

Platform: Raspberry Pi 5 + Hailo AI HAT+
AI Accelerator: Hailo-8L (13 TOPS)
Program: MSc Internet of Things and Smart Systems

1. Use Case and Problem Statement

Edge artificial intelligence is no longer confined to controlled laboratory environments; it is actively shaping high-stakes, real-world applications such as autonomous navigation, intelligent surveillance, and assistive robotics. However, as we increasingly deploy deep learning models directly onto embedded devices at the network edge, a glaring vulnerability has emerged: our current systems are dangerously reliant on single-modality perception, predominantly vision.

While camera-based sensing provides profound semantic richness—allowing a machine to meaningfully "understand" its surroundings—it is inherently fragile in the wild. When confronted with adverse environmental realities such as sudden motion blur, physical occlusions, or low-light conditions, the performance of purely visual systems deteriorates rapidly.

To counteract these visual blind spots, LiDAR presents a robust complementary modality. Because it maps the world using geometric and distance measurements, LiDAR remains steadfastly reliable regardless of ambient lighting. Yet, LiDAR is not a panacea; it is semantically blind and frequently compromised by highly reflective or absorptive surfaces. We are therefore left with a profound dichotomy: cameras understand *what* is in the environment but fail in poor conditions, while LiDAR understands *where* things are but lacks context and struggles with complex materials.

These complementary strengths and weaknesses expose a critical limitation in contemporary edge AI deployments. Currently, there is a distinct absence of robust, intelligent mechanisms designed to handle sensor degradation or outright hardware failure. In critical real-world deployments, a system cannot simply stop functioning when a camera is blinded by the sun, or a LiDAR pulse is scattered by glass.

This project directly confronts this fragility by investigating how vision–LiDAR fusion can serve as the foundation for a deeply resilient edge AI system. By integrating these modalities, this research explores how intelligent redundancy, confidence-based decision fusion, and adaptive fallback strategies can allow a system to maintain reliable perception even when facing partial sensor failure.

Ultimately, this study challenges the prevailing paradigm in the field. The central objective is to move beyond the narrow pursuit of performance optimisation under ideal conditions and instead establish a rigorous framework for *failure-aware edge intelligence*—all while operating within the strict computational and power constraints inherent to resource-limited embedded hardware.

2. Data Strategy

The evaluation combines offline benchmarking with real-time data acquisition. For controlled experimentation, the MS COCO 2017 validation dataset is used to assess object detection performance. This dataset provides a standard reference for evaluating accuracy using established metrics such as mean average precision (mAP).

To simulate realistic operating conditions, synthetic perturbations are applied to the dataset. These include Gaussian noise, motion blur, and variations in brightness. The purpose is to assess how model performance degrades under controlled forms of visual distortion.

In addition to static data, live image streams are captured using the Raspberry Pi Camera Module 3. This allows measurement of end-to-end latency under real operating conditions. In parallel, the Hokuyo URG-04LX-UG-01 LiDAR provides continuous range measurements.

To enable fusion, LiDAR data is transformed from polar coordinates into Cartesian space and aligned with the camera frame. Controlled failure scenarios are introduced by degrading one modality at a time, for example, by reducing illumination for the camera or injecting noise into LiDAR scans. This approach allows systematic evaluation of system behaviour under partial sensor failure.

3. Sensing Concept

3.1 Primary Sensor

The system combines a camera and a LiDAR sensor to provide complementary information. The Raspberry Pi Camera Module 3 captures RGB frames at a resolution of 640 × 640 pixels, with a target frame rate of up to 30 frames per second. This provides the primary source of semantic information.

The Hokuyo URG-04LX-UG-01 LiDAR provides two-dimensional range measurements. These measurements are independent of lighting conditions and support spatial understanding of the environment.

3.2 Spatial and Temporal Alignment

To integrate these modalities, LiDAR data is converted from polar to Cartesian coordinates. This enables alignment with the image plane. Temporal synchronisation is applied to ensure that each LiDAR scan corresponds to the correct image frame. Calibration procedures are used to maintain consistent alignment between the sensors.

3.3 Fusion Strategy

The system employs a fusion mechanism that operates at both feature and decision levels. Outputs from the vision model and LiDAR processing are combined using confidence-based weighting. When one modality becomes unreliable, the system adjusts its reliance on the remaining sensor. This enables continued operation under partial failure conditions.

4. System Architecture and Pre-processing

The system architecture follows a structured pipeline consisting of data acquisition, pre-processing, inference, and post-processing.

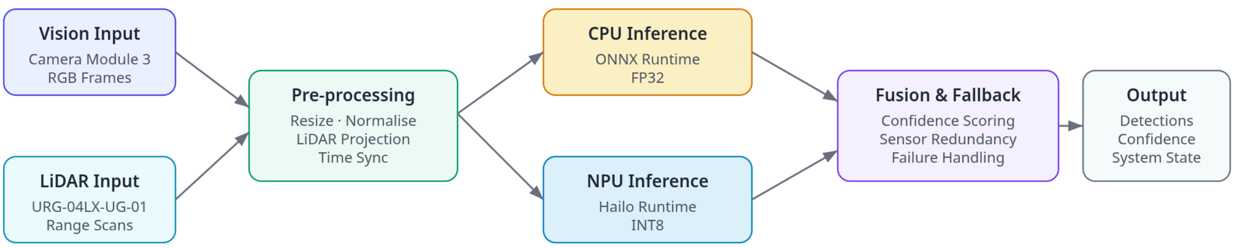

Figure 1: Edge AI inference pipeline

Pre-processing includes image resizing, normalisation, LiDAR projection, and temporal alignment. For the NPU pathway, the YOLOv8n model is quantised to INT8 using the Hailo Dataflow Compiler. This step is required to match the computational format supported by the accelerator.

Inference is performed using two execution paths. The CPU baseline uses ONNX Runtime with FP32 precision. The accelerated path uses the Hailo runtime with an INT8 model compiled into Hailo Executable Format (HEF).

Post-processing includes non-maximum suppression, bounding box decoding, and fusion of outputs from both sensing modalities. The system also includes a logging mechanism that records latency, throughput, power consumption, and temperature for each run.

5. Machine Learning Model

The system uses the YOLOv8n object detection model due to its relatively low computational complexity and suitability for embedded deployment. The model provides a balance between accuracy and efficiency.

Two versions of the model are deployed. The CPU implementation uses a full-precision FP32 model. The NPU implementation uses an INT8-quantised version optimised for the Hailo accelerator.

The outputs of the model are combined with LiDAR-derived spatial information. A confidence-based mechanism is used to integrate predictions from both modalities. This allows the system to maintain detection capability when one sensor is degraded.

6. Evaluation Metrics

The evaluation considers both performance and robustness.

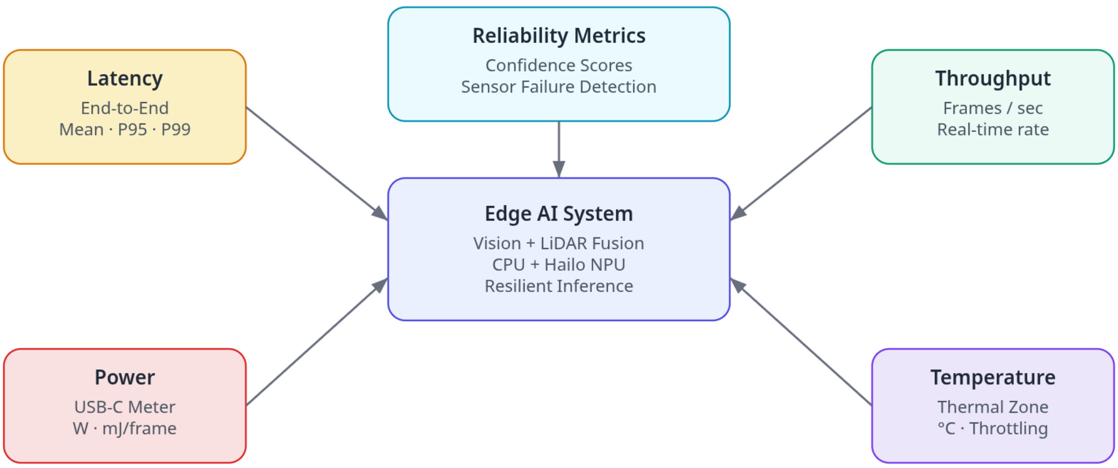

Figure 2: Benchmarking framework

Throughput is measured in frames per second to assess real-time capability. Latency is measured per frame, with additional analysis using percentile metrics to capture variability.

Power consumption is measured using an external USB-C inline meter. Energy per inference is calculated using:

$$E = P \times t.$$

Accuracy is evaluated using standard object detection metrics derived from the COCO evaluation protocol.

To assess robustness, additional metrics are introduced. These include performance degradation under sensor failure, recovery behaviour after degradation, and the effectiveness of the fusion mechanism in maintaining detection accuracy.

System-level metrics, including CPU utilisation, memory usage, and temperature, are also monitored to provide a complete performance profile.

7. Limitations and Mitigation Strategies

Sensor calibration is a critical requirement for accurate fusion. Misalignment between camera and LiDAR data can reduce system performance. This is addressed through calibration procedures and validation checks.

LiDAR data is inherently sparse, which may limit performance in complex environments. This is mitigated through interpolation and projection techniques.

Dependence on the Hailo toolchain introduces constraints related to vendor-specific software. To reduce this risk, model behaviour is validated independently before deployment on the accelerator.

Thermal constraints may affect performance under sustained workloads. Active cooling and continuous temperature monitoring are used to mitigate this issue.

8. Feasibility Assessment

Despite the ambitious scope of engineering a deeply resilient, multi-modal edge AI architecture, the practical execution of this research is fundamentally sound within the bounds of our available resources and operational timeline.

Our hardware strategy deliberately grounds this ambition in proven, accessible technology. The Raspberry Pi 5 delivers a formidable baseline computational backbone, while the Hailo AI HAT+ provides a highly stable, mathematically mature ecosystem for dedicated NPU acceleration. Furthermore, the physical integration of our geometric sensing is significantly de-risked by the selection of the Hokuyo URG-04LX-UG-01 LiDAR. Because this sensor is backed by robust, well-documented interfaces and highly stable drivers, we effectively neutralise the notoriously complex, low-level hardware integration hurdles that often paralyse multi-modal computing projects.

Crucially, **by intentionally stripping away any reliance on traditional, power-hungry GPU acceleration, we actively eliminate a massive source of systemic and thermal instability**. This deliberate architectural constraint not only aligns with the harsh realities of true edge computing, but it guarantees a tightly controlled, empirically rigorous environment for our core baseline experiments.

Ultimately, the true intellectual and engineering friction of this project lies in the successful synthesis of our divergent data streams—specifically, the execution of robust, adaptive sensor fusion. However, this primary challenge remains highly manageable; by committing to a rigorous philosophy of incremental development and continuous algorithmic validation, we can systematically construct and prove the resilience of this multi-modal framework without jeopardising the project's fundamental viability

9. Expected Outcomes

The NPU-based implementation is expected to provide improved inference latency and energy efficiency compared to the CPU baseline.

The fusion approach is expected to improve system robustness under sensor degradation. The study will quantify how performance is maintained when one modality becomes unreliable.

The outcome will be a framework for designing edge AI systems that can operate under partial sensor failure while maintaining acceptable performance.

10. Conclusion

This project investigates the design of a resilient edge AI system using vision–LiDAR fusion and hardware acceleration. It addresses the limitations of single-modality perception and evaluates the role of NPUs in improving efficiency on embedded platforms.

The results are expected to provide practical insights into the design of robust AIoT systems that can operate reliably under real-world conditions.